Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	GEEK FAM
Project / Solution Name	AI CAREER NAVIGATOR
Name (Leader)	MUHAMMAD AMMAR HARIZ BIN YAZID HAMDI
Email & Phone (Leader)	ammarhariz140807@gmail.com/01139060828
University Affiliation(s)	UNIVERSITI TEKNIKAL MALAYSIA MELAKA (UTeM)
Live Prototype URL	AI Career Navigator
Code Repository URL (public / access granted to judges)S	https://github.com/admirfn988/Career-Navigation.git

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

AI Career Navigator is a three-tier web application hosted on Replit. The user interface is a React and TypeScript single-page app (Vite, TanStack Query) where students enter their education, skills and interests, then explore career matches, skills gaps, learning roadmaps, SDG alignment and AI-disruption scenarios. Matching and gap analysis run in the browser against a curated career dataset, so results appear instantly. The interface talks to a Flask REST API (/api/career/state) over same-origin HTTPS. The backend validates every request and identifies each visitor through a signed, HTTP-only session cookie. It stores the profile, selected career and scenario in an external MySQL database, so progress survives page reloads. API contracts are defined once in an OpenAPI specification, which generates the typed client hooks and Zod schemas shared by frontend and backend. A lightweight Express service provides health checks. End-to-end flow: user interface → Flask API → MySQL, with Replit hosting all services.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
React 19.1.0	Frontend UI framework	MIT
TypeScript 5.9	Typed language for the frontend and Node code	Apache-2.0
Vite 7 + Tailwind CSS 4	Frontend build tool and styling	MIT
TanStack React Query 5 + wouter 3	API data fetching and page routing	MIT
Radix UI / shadcn/ui, Recharts 2, Framer Motion 12, Lucide React 0.545	UI components, charts, animations and icons	MIT (Lucide: ISC)

Name & Version	Purpose in Project	License
Python 3.13	Language for the Flask backend	PSF License
Flask 3.1.3 + PyMySQL 1.2.3 + Gunicorn 26.2	Backend REST API, MySQL connection, production server	BSD-3-Clause (Flask), MIT (PyMySQL), MIT (Gunicorn)
Orval 8	Generates API client code from the OpenAPI spec	MIT
Drizzle ORM 0.45	Database schema definitions	Apache-2.0

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Hosts and deploys the prototype (React frontend, Flask API, Express health service) and provides the development environment	Replit Core, sponsor-provided (free for the hackathon) / Region managed by Replit
MySQL database	Stores user profile, selected career and scenario, linked to a session ID	
Replit Secrets	Stores the session secret and database connection string outside the code	Included with Replit Core

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Career catalogue (6 sample careers with skills, interests, outlook)	Created by our team	Self-collected / sample	Own work	N
Skill and interest option lists	Created by our team	Self-collected	Own wor	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Learning resource links and SDG references	Public sites: Coursera, edX, Microsoft Learn, Kaggle, Google Cloud Skills Boost, WEF Future of Jobs 2025, BLS, UN SDGs (static links only, no data downloaded)	Public	Not applicable	N
User profile (education, skills, experience, projects, interests	Entered by users in the app, stored in MySQL	Private (user-provided)	Not applicable	Y

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Replit Agent / Replit AI	Project setup, generating and editing the React frontend and Flask backend code	[major / moderate]
Claude (Anthropic)	Debugging, code review, drafting submission documents, and guidance on using Git to upload project files	[moderate]
Gemini (Google)	Ideas, code help, and helping create the presentation slide deck	[minor / moderate]

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

All AI-generated code, text and slides were reviewed by team members before use. We ran the application, tested each page and the save and reload flow, and fixed errors found during testing. We checked that library versions, licences and the features described in this document match the actual code. AI suggestions were accepted, edited or rejected by the team. No API keys or secrets were shared in AI prompts or stored in the repository. The team takes full responsibility for the final submission.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Replit Core plan (platform and credits)	Used as the development environment and hosting platform for the React frontend, Flask API and Express service

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

The sponsor resource was useful but not essential to the core functionality. Replit Core gave us a convenient environment for building, running and hosting the prototype, and it let us work faster. The application itself is standard web code (React, Flask, MySQL), so it could be moved to another hosting platform with minor configuration changes. The career matching logic does not depend on any sponsor API or model.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The application does not require accounts and does not collect names, emails or contact details. Users enter their education, skills, experience, projects and interests, which can be personal information. This profile, the selected career and the scenario are stored in an external MySQL database, linked only to a random session ID kept in a signed, HTTP-only cookie. No user data is sent to any AI model or other third-party API. Data is protected by server-side input validation and length limits, parameterised database queries, encrypted database connections (TLS), and secrets kept outside the code in Replit Secrets. Currently, data is kept until the database is cleared, because we have not yet built an automatic deletion feature. Only team members with database access can view stored data.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

The prototype has no AI model, so it has no hallucination risk, but it has other limits. Career matching is rule-based and uses a small sample catalogue of six careers, so results are simple guidance, not professional advice. Disruption scenarios are illustrative and not precise predictions. Matching may be biased toward the roles and skills we chose to include. The app depends on a single hosting provider (Replit) and a single database, so an outage makes saving unavailable (the API reports this clearly and never shows a failed save as successful). Sessions are browser-based, so clearing cookies loses access to a saved profile. Retention limits and a delete-my-data option are planned but not implemented. As mitigation, we validate all input, label sample data, and link users to official sources.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.

☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.

☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.

☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.

☑ Any use of official SDC sponsor resources has been declared in Section 5.

☑ The prototype described here is accessible to judges through the links provided in Section 1.

☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



[MUHAMMAD AMMAR HARIZ BIN YAZID HAMDI]

[24/11/2026]